

2013-2014 – Master 2 – Macro I

Lecture 5 : Sustained Growth

Franck Portier

(based on Gilles Saint-Paul lecture notes)

franck.portier@TSE-fr.eu

Toulouse School of Economics

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Disclaimer

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0. Introduction

- ▶ We want to understand the conditions under which sustained growth may occur in the absence of an exogenous trend of technical progress.
- ▶ For this purpose we start with a fairly general growth model.
- ▶ We have accumulable factors such as physical and human capital and fixed factors such as labor and land.
- ▶ We want to know if growth can come from the sheer accumulation of the first type of factors.
- ▶ Sustained growth is often referred to as *endogenous growth*.

1. A general growth model

Technology

- ▶ p accumulable factors indexed by $i = 1, \dots, p$ and denoted by K_i
- ▶ n non accumulable factors indexed by $j = 1, \dots, n$ and denoted by L_j .
- ▶ There is a production function which tells us the maximum rate at which each factor can be accumulated :

$$\dot{K}_i \leq G_i(K_1, \dots, K_p, L_1, \dots, L_n). \quad (1)$$

- ▶ **Remark 1 :** We do not need to specify the utility function in order to examine whether sustained growth is possible. It depends on technology only.
- ▶ **Remark 2 :** The amount left $G_i - \dot{K}_i$ may be positive if it enters the utility function. Otherwise it is optimal that the above constraint be binding.

1. A general growth model

Technology (continued)

$$\dot{K}_i \leq G_i(K_1, \dots, K_p, L_1, \dots, L_n). \quad (1)$$

- ▶ **Remark 3 :** Typically we may have $\dot{K}_i = G_i(s_{i1}K_1, \dots, s_{ip}K_p, \sigma_{i1}L_1, \dots, \sigma_{in}L_n)$ where s_{ij} (resp. σ_{ij}) is the share of accumulable factor j (resp. non-accumulable factor j) employed in the production of accumulable factor i . Since s and σ are between 0 and 1 and G is increasing, (1) clearly holds.
- ▶ **Example :** In the standard Ramsey growth model $p = n = 1$ and G_1 is just the production function for output net of depreciation, $G_1 = F(K, L) - \delta K$. The difference $G_1 - \dot{K}$ is the flow of consumption. In this model, the final good can be used for either consumption or capital accumulation. This need not be true for all models and all accumulable factors.

1. A general growth model

Condition for sustained growth

- ▶ Let

$$\eta_{ij} = \frac{K_j}{G_i} \frac{\partial G_i}{\partial K_j}$$

and

$$q_i = \sum_{j=1}^p \eta_{ij}.$$

- ▶ Note that both the η_{ij} s and the q_i s need not be constant : They depend on the input mix and can change over time. However, in what follows I will assume that in a BGP these quantities have a limit as time goes to infinity. This is typically the case.
- ▶ Consider a balanced growth path such that $\dot{K}_j/K_j = g_j$.
- ▶ This BGP has sustained growth provided $\exists i, g_i > 0$.

1. A general growth model

Condition for sustained growth (continued)

- Assume this is the case and let $\bar{g} = \max_j g_j > 0$ and let i such that $g_i = \bar{g}$. We must have for all j .

$$0 \leq g_j \leq \frac{G_j}{K_j} \quad (2)$$

and for i :

$$0 < g_i \leq \frac{G_i}{K_i} \quad (2b)$$

- Next,

$$\begin{aligned} \frac{d \log \left(\frac{G_i}{K_i} \right)}{dt} &= \frac{\dot{G}_i}{G_i} - \frac{\dot{K}_i}{K_i} = \sum_{j=1}^p \eta_{ij} g_j - g_i \\ &\leq (q_i - 1) \bar{g}. \end{aligned}$$

- Suppose $\lim_{t \rightarrow \infty} q_i < 1$. Then there exists some upper bound $\bar{q}$ such that $q_i < \bar{q} < 1$ for t large enough. Then $\lim_{t \rightarrow \infty} \frac{G_i}{K_i} = 0$ which contradicts (2b)

1. A general growth model

Constant returns in accumulable factors

- ▶ We therefore have
- ▶ **Theorem 1 :** For sustained growth to be possible there must exist i such that $\lim q_i \geq 1$. Furthermore,
 $i \in \arg \max_i g_i \Rightarrow \lim q_i \geq 1$.
- ▶ Thus, *asymptotically*, we must have returns to accumulated factors equal to or greater than one for at least one accumulable factor of production.
- ▶ **Remark 4 :** If non accumulable factors enter G_i , this condition cannot be satisfied if there are constant returns to scale to all factors.
Remark 5 : We can get $q_i = 1$ with a standard production function provided all factors are accumulable.

1. A general growth model

A stronger result

- By noting that other factors are accumulated too, we can derive a stronger result. Take any j and assume again that $g_i = \bar{g}$. Then

$$\frac{\dot{G}_j}{G_j} - \frac{\dot{K}_j}{K_j} \leq q_j g_i - g_j. \quad (3)$$

- If $g_j > 0$ then the LHS of (3) cannot be negative, otherwise (2) would be violated. Thus it must be that

$$g_j \leq q_j \bar{g},$$

which obviously also holds if $g_j = 0$.

- This allows us to derive another inequality involving g_i :

$$\begin{aligned} \frac{\dot{G}_i}{G_i} - \frac{\dot{K}_i}{K_i} &= \sum_{j=1}^p \eta_{ij} g_j - g_i \\ &\leq \left(\sum_{j=1}^p \eta_{ij} q_j - 1 \right) \bar{g}. \end{aligned}$$

1. A general growth model

A stronger result (continued)

$$\lim_{t \rightarrow \infty} \sum_{j=1}^p \eta_{ij} q_j \geq 1. \quad (4)$$

- ▶ In general a condition like (4) is not stronger than the condition $q_i \geq 1$, except in the important "neo-classical" case where all the G s have constant returns to all their factors.
- ▶ We can then use the preceding logic to derive an interesting theorem :
- ▶ **Theorem 2** : Assume that the following holds :
 - ▶ *Assumption A1* : The model is *neoclassical*, i.e : G_i is homogeneous of degree one in all its inputs.
 - ▶ The model exhibits sustained growth, implying (4) holds.

Then let $i \in \arg \max_i g_i$. Then $\lim_{t \rightarrow \infty} q_i = 1$. Furthermore, for any j such that $\lim \eta_{ij} > 0$, one must have $\lim q_j = 1$.

1. A general growth model

Proof of theorem 2

- ▶ A1 implies that for all factors j , $q_j \leq 1$.
- ▶ Consequently,
$$\lim_{t \rightarrow \infty} \sum_{j=1}^P \eta_{ij} q_j \leq \lim_{t \rightarrow \infty} \sum_{j=1}^P \eta_{ij} = \lim_{t \rightarrow \infty} q_i.$$
- ▶ Therefore, from (4) $\lim_{t \rightarrow \infty} q_i \geq 1$.
- ▶ Since one must also have $q_i \leq 1$, it must be that
$$\lim_{t \rightarrow \infty} q_i = 1.$$
- ▶ Furthermore, let j_0 such that $\lim \eta_{ij_0} > 0$. WLOG, take $j_0 = 1$.
- ▶ Then $\lim_{t \rightarrow \infty} \sum_{j=1}^P \eta_{ij} q_j = \lim \sum_{j=2}^P \eta_{ij} q_j + \lim \eta_{i1} q_1 \leq$
$$\lim \sum_{j=1}^P \eta_{ij} + \lim \eta_{i1} (q_1 - 1) = \lim q_i + \lim \eta_{i1} (q_1 - 1) =$$
$$1 + \lim \eta_{i1} (q_1 - 1).$$
- ▶ Since this must also be ≥ 1 , it must be that
$$0 \leq \lim \eta_{i1} (q_1 - 1) = \lim \eta_{i1} \lim (q_1 - 1) \implies \lim q_1 \geq 1 \implies$$
$$\lim q_1 = 1.$$
- ▶ For any accumulable input which *matters* for the production of the fastest growing good i , the production function of such inputs must asymptotically involve accumulable factors only.

2. Examples

The AK model

- ▶ The AK model is the Ramsey model where the production function is replaced with $Y = AK$.
- ▶ We then have $G_1(K, L) = AK$ and sustained growth is possible.
- ▶ **Exercise 1 :** Assume that at each date consumption is $C = (1 - s)Y$. What is the long-run growth rate of the economy ?
- ▶ **Exercise 2 :** Solve the Ramsey model with $Y = AK$ and derive the long-run growth rate.

2. Examples

The Jones-Manuelli model

- ▶ In the Jones-Manuelli model one has $Y = AK + BK^\alpha L^{1-\alpha}$.
- ▶ Here we have $q_1 = \frac{A+\alpha B(K/L)^{\alpha-1}}{A+B(K/L)^{\alpha-1}} < 1$, but in a BGP where $g_K > 0$, $\lim q_1 = 1$.
- ▶ Therefore the necessary conditions of theorem 1 are satisfied and sustained growth is indeed possible.
- ▶ **Exercise 3 :** Does sustained growth match the stylized facts of growth in the AK and Jones-Manuelli models? Which of those stylized facts are violated?
- ▶ **Exercise 4 :** Discuss if sustained growth is possible for $Y = (K^\beta + L^\beta)^{1/\beta}$.

2. Examples

The Lucas model

- ▶ In the Lucas model there are 2 accumulable factors of production, H = human capital and K = physical capital. We have

$$Y = AK_Y^\alpha H_Y^\beta L_Y^{1-\alpha-\beta}$$

$$\dot{K} = Y - C$$

$$\dot{H} = BK_H^\delta H_H^\gamma L_H^{1-\delta-\gamma}.$$

$$K_H + K_Y = K$$

$$H_H + H_Y = H$$

$$L_H + L_Y = L,$$

and L is constant through time.

- ▶ We can take $K_1 = K, K_2 = H$,
 $G_1(H, K, L) = AK^\alpha H^\beta L^{1-\alpha-\beta}, G_2(H, K, L) = BK^\delta H^\gamma L^{1-\delta-\gamma}.$
- ▶ We have $q_1 = \alpha + \beta$ and $q_2 = \delta + \gamma$. Thus, Theorem 1 implies that to have sustained growth we need $\alpha + \beta = 1$ or $\delta + \gamma = 1$.

2. Examples

The Lucas model (continued)

$$G_1(H, K, L) = AK^\alpha H^\beta L^{1-\alpha-\beta}, G_2(H, K, L) = BK^\delta H^\gamma L^{1-\delta-\gamma}.$$

- ▶ Furthermore, theorem 2 shows that there are only 3 possibilities :
 1. $\alpha, \beta, \gamma, \delta < 1$ and $\alpha + \beta = \gamma + \delta = 1$. In this case all factors are accumulable.
 2. $\alpha = 1$, i.e. human capital and labor do not matter for capital accumulation. We are then back to the AK model.
 3. $\gamma = 1$, i.e. physical capital and labor do not matter for human capital, accumulation. We could call this the BH model.